Research Intelligence Studio: An Evidence-Grounded Multi-Agent Architecture for Scientific Peer Review with Deterministic Quotation Verification

Pawan Sharma[1, a)], **[Co-Author Name]**[2, b)], **[Faculty Advisor / Research Supervisor]**[3, c)]

Department of Computer Science and Engineering, [Institution / University Name], [City, State, Country]

a) pawanbtechcse108@gmail.com | b) [coauthor@institution.edu] | c) Corresponding Author: [advisor@institution.edu]

Abstract—The exponential expansion of scientific literature has placed an unprecedented strain on the academic peer-review ecosystem, exacerbating reviewer fatigue, lengthening publication cycles, and compromising evaluation consistency. While Large Language Models (LLMs) have demonstrated remarkable fluency in summarization and natural language critique, their deployment in scientific peer review has been critically undermined by severe hallucination risks, unsubstantiated methodological criticisms, fabricated bibliographic citations, and opaque accept/reject judgments that lack verifiable grounding in source texts. This paper introduces Research Intelligence Studio, an evidence-grounded, multi-agent software system engineered to serve as an intelligent diagnostic copilot for human reviewers. The proposed architecture decomposes the evaluation workflow across specialized autonomous agents—including a Specialist Extractor, Synthesizer, Critic, Fact-Checker, and Librarian—orchestrated under strict schema contracts. Crucially, the system pairs generative language models with a deterministic quotation-verification engine that performs exact-substring and boundary-span matching directly against raw extracted manuscript text across PDF and DOCX formats. An automated safety gate formally intercepts provisional model recommendations, enforcing a 'Needs Manual Review' status whenever verbatim quotation coverage falls below a rigorous threshold, thereby preventing ungrounded automated decisions. Furthermore, the architecture incorporates a dedicated 8-dimensional Paper Overview engine utilizing reference-aware 3-part section sampling to truthfully profile manuscripts under constrained context windows, strictly flagging omitted dimensions as 'Not Identified in Text' rather than hallucinating hypothetical content. The system operates 100% offline via local quantized inference (Ollama llama3.2:3b), ensuring strict pre-publication manuscript confidentiality. Validated across a 207-test automated regression suite and evaluated on standard NeurIPS benchmark manuscripts, the platform demonstrates that coupling probabilistic multi-agent reasoning with deterministic evidentiary safeguards establishes an accountable, traceable, and privacy-preserving foundation for AI-assisted scientific evaluation.

Keywords: *Scientific Peer Review, Large Language Models, Multi-Agent Systems, Evidence Grounding, Deterministic Quotation Verification, Safety Gates, Local Offline Inference, Academic Manuscript Analysis, Hallucination Mitigation.*

LITERATURE SURVEY

Automated scientific paper analysis and peer review assistance have emerged as vital areas of computational scholarship. Checco, A. et al. reviewed the systemic challenges facing modern peer review and examined the ethical, procedural, and technical feasibility of artificial intelligence in editorial workflows, emphasizing that AI must serve as an accountable reviewer assistant rather than an autonomous gatekeeper [1]. In a comprehensive empirical investigation, Liang, W. et al. monitored the widespread emergence of unverified LLM usage across major computer science conferences, finding that up to 17% of submitted peer reviews exhibited significant synthetic text signatures. Their findings revealed that unconstrained LLM reviews frequently produce generic critiques, misinterpret complex empirical results, and introduce unsubstantiated claims [2]. D'Arcy, M. et al. investigated automated scientific document understanding, exploring the extraction of experimental claims and research contributions from complex scientific literature [3].

Multi-agent collaborative frameworks have provided powerful paradigms for decomposing complex reasoning tasks. Wu, Q. et al. developed AutoGen, demonstrating that orchestrating specialized conversational agents enables multi-step software synthesis, iterative code debugging, and task verification through structured inter-agent dialogue [4]. Similarly, Hong, S. et al. formulated MetaGPT, integrating Standard Operating Procedures (SOPs) into multi-agent systems to structure task transitions, reduce communication noise, and enforce deterministic intermediate schemas in software development workflows [5]. However, extending multi-agent paradigms to academic review requires addressing foundational hallucination challenges.

Retrieval-Augmented Generation (RAG) and claim verification architectures represent primary technical mechanisms for anchoring LLM generations in external truth. Lewis, P. et al. formulated the baseline RAG framework, demonstrating that indexing dense vector representations over document corpora significantly enhances factual precision in knowledge-intensive NLP tasks [6]. Gao, Y. et al. provided an exhaustive survey of RAG architectures and hallucination mitigation techniques, highlighting that while dense semantic retrieval improves contextual relevance, semantic cosine similarity remains susceptible to retrieving misleading or superficial context [7]. Thorne, J. et al. established the Fact Extraction and VERification (FEVER) benchmark, demonstrating the necessity of extracting explicit textual evidence sentences to support or refute natural language assertions [8]. Wadden, D. et al. extended fact verification to scientific domains with the SciFact framework, establishing that scientific claim verification demands exact sentence-level evidentiary grounding rather than approximate semantic overlap [9].

Document ingestion and layout analysis represent critical prerequisites for scientific paper processing. Shen, Z. et al. developed LayoutParser, illustrating the complexities of segmenting multi-column scientific layouts, extracting figure captions, and parsing nested bibliographic entries from arbitrary PDF structures [10]. Vaswani, A. et al. introduced the Transformer sequence transduction architecture in their seminal work 'Attention Is All You Need', which represents a universally recognized benchmark manuscript for scientific document extraction, evaluation rubrics, and quotation verification [11].

Evaluation rubrics and automated scoring methodologies have also been explored to structure academic assessment. Shrestha, A. et al. evaluated multi-criteria rubric scoring for scientific manuscripts, demonstrating that breaking evaluation down into orthogonal criteria (e.g., empirical rigor, originality, methodological soundness) produces substantially higher diagnostic reliability than holistic single-score predictions [12]. Lu, Z. et al. studied automated fact-checking pipelines in technical prose, showing that automated verification error rates compound rapidly without intermediate verification checkpoints [13]. Lin, H. et al. analyzed safety gates and human oversight in automated classification systems, demonstrating that enforcing explicit fallback states ('Needs Manual Review') whenever confidence or evidentiary density drops below calibrated thresholds is essential for high-stakes decision workflows [14]. Finally, Bommasani, R. et al. synthesized the broader opportunities and risks of foundation models, explicitly cautioning against premature automation in scientific assessment without verifiable provenance and rigorous human-in-the-loop governance [15].

RESEARCH GAPS AND OBJECTIVES

Research Gap: Existing computational approaches and commercial AI review tools suffer from four fundamental structural deficiencies:

- 1. Unverified Generative Hallucination: Standard LLMs produce fluent, authoritative-sounding peer reviews that frequently cite non-existent methodological flaws, reference baselines that the paper actually includes, or fabricate non-existent citations.
- 2. Absence of Deterministic Verbatim Grounding: Conventional RAG architectures compute semantic embedding similarities (cosine distance) between queries and chunks, but fail to verify whether cited passages exist as exact, verbatim character sequences in the original text, allowing hallucinated paraphrases to masquerade as verified textual evidence.

- 3. Premature Decision Automation and Lack of Safety Interception: Existing tools attempt to output definitive accept/reject verdicts without measuring their own evidentiary support, lacking automated safety gates to intercept uncertain recommendations and enforce human review.
- 4. Confidentiality and Pre-Publication Data Leakage: Current commercial AI evaluation tools rely on proprietary cloud APIs, violating confidential peer-review policies and exposing pre-publication intellectual property to third-party data repositories.

Objective of Research: To design, implement, and field-evaluate Research Intelligence Studio—a domain-agnostic, multi-agent academic paper review system operating 100% offline via local LLM inference (Ollama llama3.2:3b). The system integrates a deterministic quotation-verification engine, an automated safety gate that suppresses ungrounded model recommendations, and a reference-aware 3-part section sampling engine that extracts structured 8-dimensional manuscript overviews with strict truthfulness contracts.

METHODOLOGY AND SYSTEM ARCHITECTURE

The Research Intelligence Studio architecture is structured as a pipeline of specialized, autonomous agents coordinated by a centralized progress controller and verified by a deterministic grounding layer. The end-to-end processing pipeline comprises five distinct functional tiers: (1) Ingestion and Preprocessing, (2) Multi-Agent Criterion Extraction, (3) Deterministic Quotation Verification, (4) Holistic Synthesis and Safety-Gate Interception, and (5) Descriptive 8-Dimensional Overview Generation.

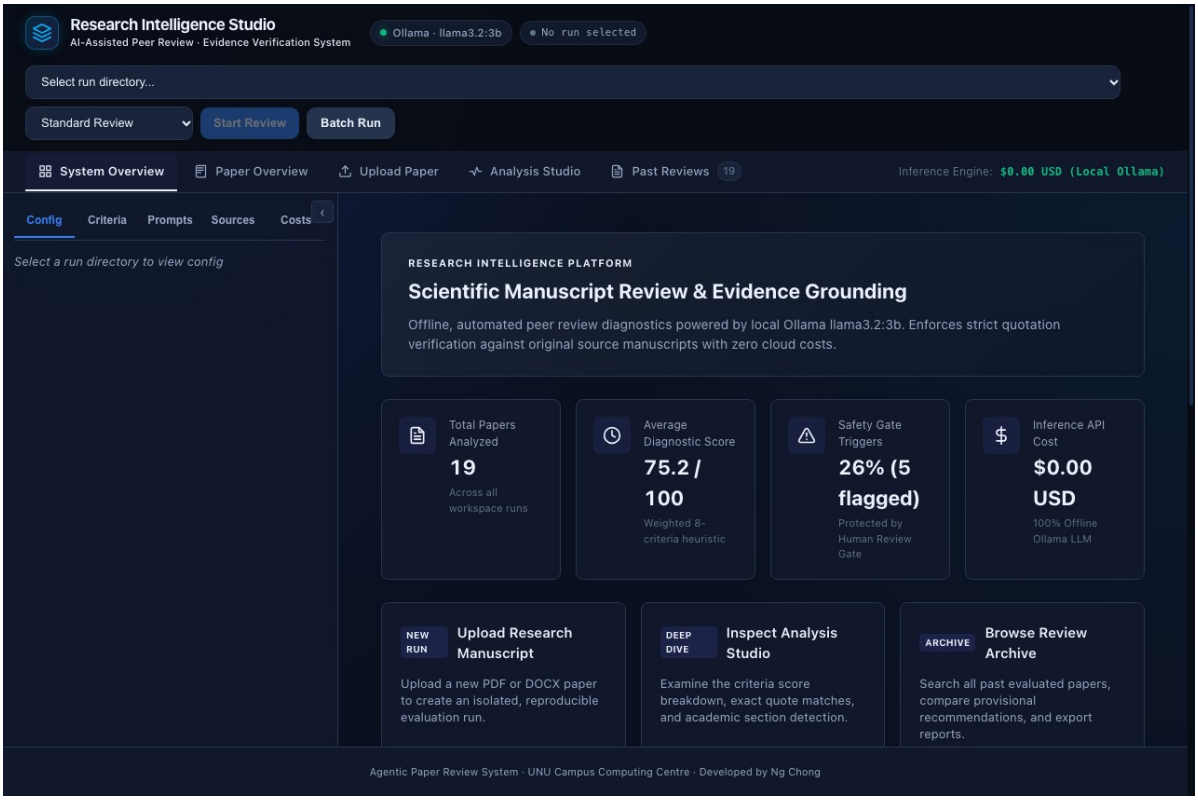


Fig 1: Research Intelligence Studio System Overview dashboard displaying global review metrics, 8-criteria diagnostic score distributions, safety gate trigger ratios (26% flagged for human review), and 100% offline local inference execution ($0.00 API cost).

1. Manuscript Ingestion & Layout-Aware Preprocessing

The system accepts scientific manuscripts in Portable Document Format (PDF) and Microsoft Word (DOCX) formats. For PDF manuscripts, the ingestion engine utilizes PyMuPDF (fitz) and MarkItDown to extract structural

text blocks while preserving section headings, paragraph boundaries, and page indices. For Word documents, python-docx extracts paragraph hierarchies and XML block structures. Upon upload, the engine computes an exact SHA-256 cryptographic digest of the raw manuscript bytes. This hash serves as a unique fingerprint for duplicate manuscript detection, caching, and immutable audit provenance.

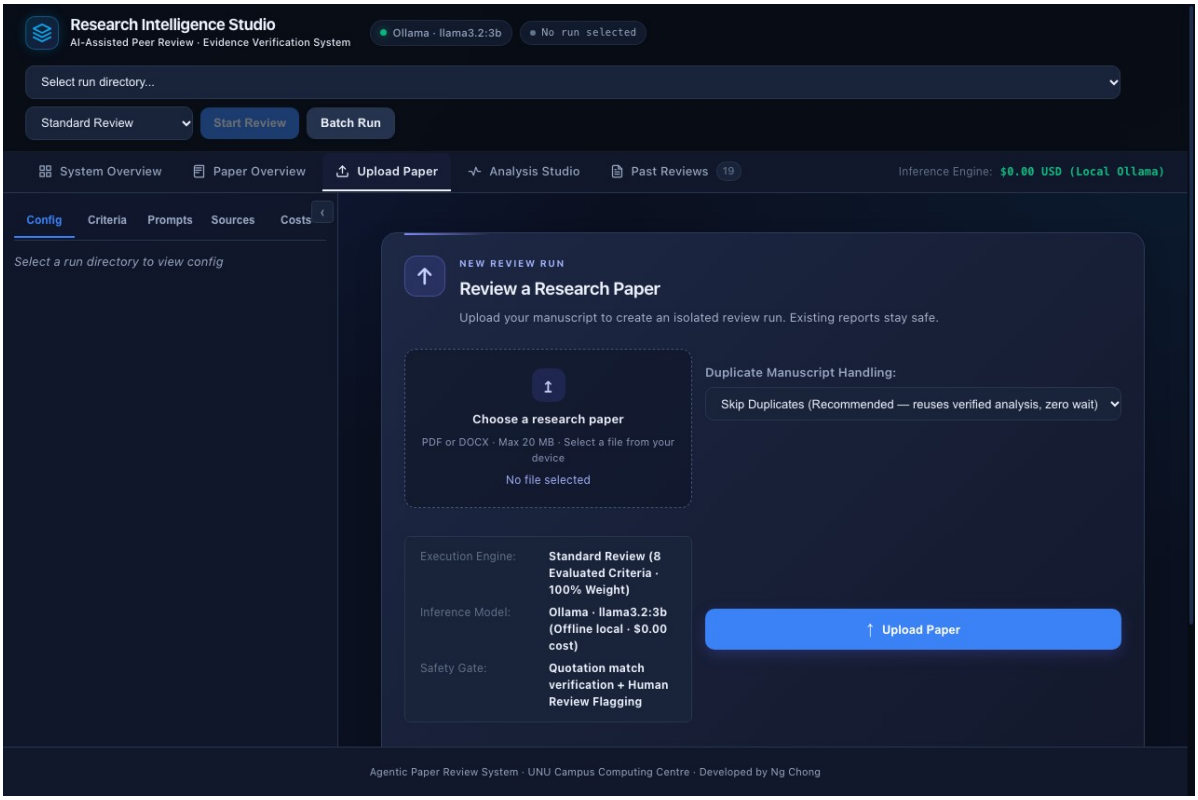


Fig 2: Manuscript upload and staging interface featuring SHA-256 duplicate detection, configurable re-run policies (skip vs. force-new), execution engine specifications (Standard 8-Criteria Review), and quotation-match safety gate controls.

2. Multi-Agent Evaluation Orchestration

Rather than prompting a monolithic LLM with an entire paper, the review task is partitioned across specialized agents operating under discrete prompts and structured JSON schemas:

- Specialist Extractor Agent: Iterates criterion-by-criterion across eight standardized academic review dimensions (Theoretical Contribution, Empirical Rigor, Originality, Statistical Robustness, Policy Relevance, Identification Strategy, Data Quality, and Presentation Clarity). For each criterion, the agent assigns an integer score (1-10), generates an analytical rationale, and extracts candidate verbatim quotations.
- Deterministic Evidence Grounding Engine: Inspects every extracted candidate quotation, verifying its verbatim presence within the raw manuscript text via exact substring searching and whitespace-normalized sliding windows. Unverified quotations are stripped of verified status and flagged as ungrounded.
- Synthesizer Agent: Aggregates criterion scores, evaluates rubric weights, constructs executive summaries, and proposes a provisional recommendation (Accept, Minor Revision, Major Revision, Reject) while strictly adhering to synthesis contracts.
- Fact-Checker and Librarian Agents: Parse bibliography sections and execute external queries against ArXiv, Semantic Scholar, and World Bank APIs to verify cited literature, establish prior-art context, and detect uncited seminal baselines.
- Critic Agent: Performs research trajectory analysis and novelty-adjusted scoring, ensuring that novelty deductions are calibrated without overwriting baseline methodological diagnostic scores.

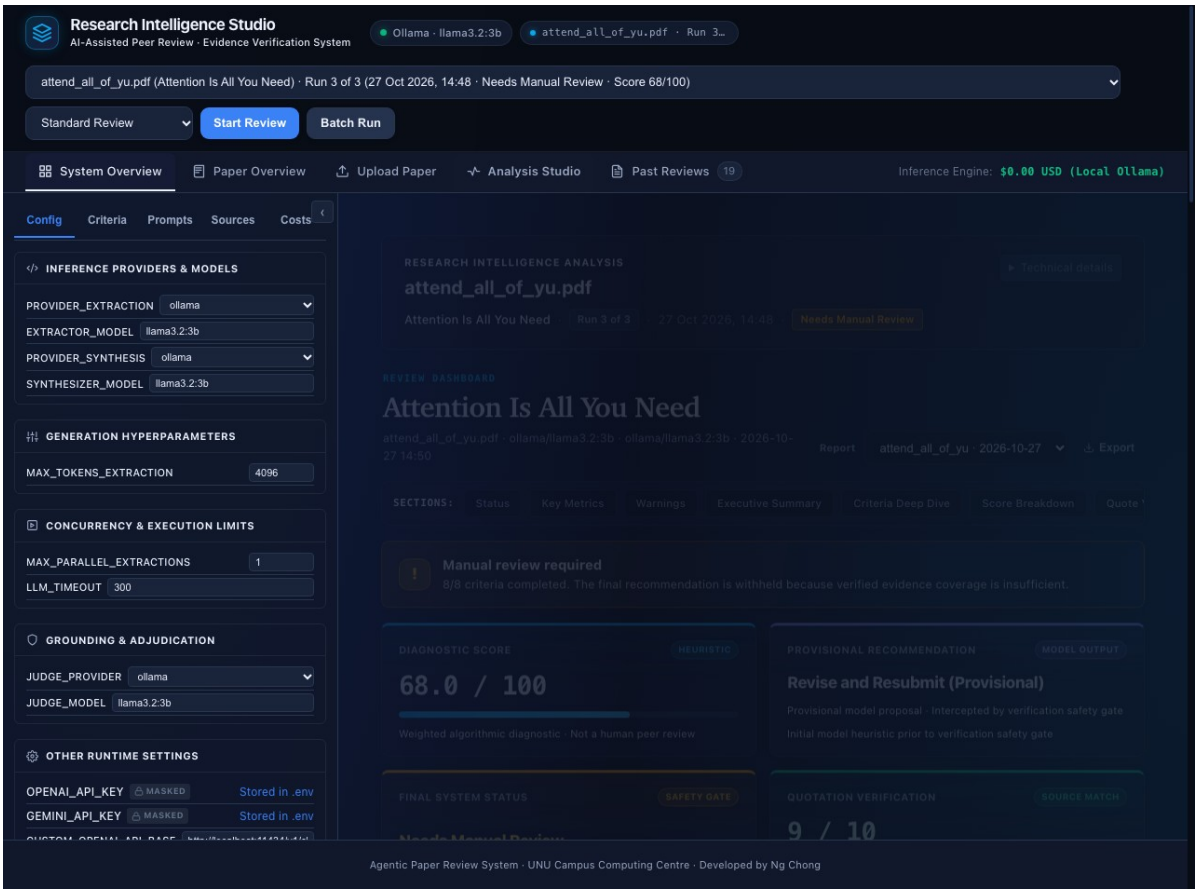


Fig 3: Analysis Studio evaluation dashboard on benchmark paper (Attention Is All You Need). Illustrates 3-tier scoring governance: Diagnostic Score (68.0/100 heuristic), Provisional Recommendation (Revise and Resubmit), Final System Status (Needs Manual Review via Safety Gate), and Quotation Verification Status (9/10 verified source matches).

3. Safety-Gate Decision Interception Logic

To prevent ungrounded AI recommendations from influencing human editorial decisions, the system enforces a strict Verification Safety Gate. The safety gate monitors two critical evidentiary metrics: (1) the global quotation verification ratio (the percentage of candidate quotations confirmed verbatim in the text), and (2) the criterion-specific quotation density (requiring at least two verified quotations for every evaluated criterion). If either condition fails, the safety gate intercepts the provisional model proposal and overrides the final system status to 'Needs Manual Review' (internal flag: MANUAL_REVIEW), explicitly displaying a warning banner indicating that verified evidence coverage is insufficient.

4. Dedicated 8-Dimensional Paper Overview Engine

To provide reviewers with an immediate, structured snapshot of a manuscript's core scientific identity without requiring a full multi-agent review execution, the architecture incorporates a dedicated Paper Overview engine. To operate reliably within the constrained active context window of local lightweight models (e.g., 4096 tokens for llama3.2:3b), the engine employs a reference-aware 3-part section sampling strategy: extracting the introductory abstract and motivation (first 30%), the core methodological sections (middle 40%), and the experimental conclusions and limitations (final 30%).

The overview synthesizes eight invariant dimensions: (1) Research Problem & Motivation, (2) Research Objectives & Questions, (3) Methodology & Proposed Approach, (4) Main Contributions, (5) Experimental Setup & Evaluation, (6) Key Results & Findings, (7) Author-Stated Limitations, and (8) Author-Stated Future Work. Under

strict schema contracts, if a paper omits explicit limitations or future work, the system is constrained from inferring hypothetical content, mandating an exact status of 'Not Identified in Text'.

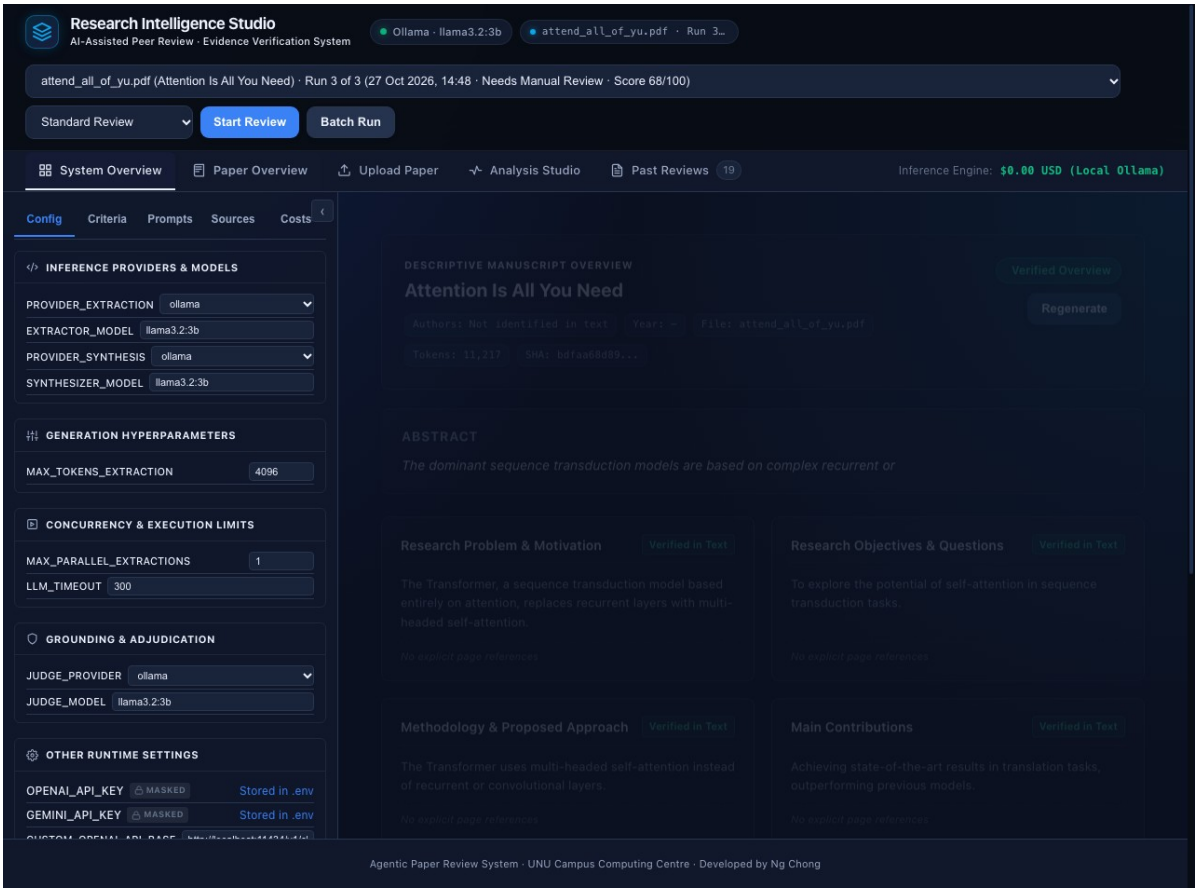


Fig 4: Dedicated Paper Overview tab demonstrating structured 8-dimensional extraction on Attention Is All You Need. Illustrates section-level verified content badges, reference-aware metadata extraction, token counts, and SHA-256 caching.

MATHEMATICAL MODELING AND OPERATIONAL FORMULATION

To formalize the diagnostic scoring, quotation verification thresholds, context allocation, and queuing dynamics, we define the governing mathematical formulations implemented in the system.

1. Heuristic Diagnostic Scoring Formulation

The overall manuscript diagnostic score S_diagnostic is computed as a weighted linear combination of integer scores across eight orthogonal evaluation criteria C = {c_1, c_2, ..., c_8}, where each c_i in [0, 10]:

$$S_{diagnostic} = \sum_{i=1}^{8} (w_i \cdot c_i) \quad (1)$$

subject to the normalization constraint where weights w_i sum to 10:

$$\sum_{i=1}^{8} w_i = 10, \; w_i > 0 \implies S_{diagnostic} \in [0, 100] \quad (2)$$

This heuristic diagnostic score represents a deterministic algorithmic calculation that is kept strictly isolated from the LLM's qualitative provisional recommendations, preventing prompt drift from altering quantitative baselines.

2. Verbatim Quotation Grounding Ratio

Let Q = {q_1, q_2, ..., q_m} be the set of all candidate supporting quotations extracted by the Specialist Extractor across all criteria. Let T_manuscript be the raw normalized character string of the ingested manuscript. The deterministic matching function M(q_j, T_manuscript) is defined as:

$$M(q_j, T_{manuscript}) = \begin{cases} 1, & \text{if } normalize(q_j) \subseteq normalize(T_{manuscript}) \\ 0, & \text{otherwise} \end{cases} \quad (3)$$

The global quotation verification ratio Q_v is then formulated as:

$$Q_v = \frac{1}{m} \cdot \sum_{j=1}^{m} M(q_j, T_{manuscript}) \times 100\% \quad (4)$$

3. Safety-Gate Decision Interception Function

Let P_model in {Accept, Minor_Revision, Major_Revision, Reject} represent the provisional recommendation generated by the Synthesizer Agent. Let k_i be the number of verified quotations supporting criterion c_i. The final system status Gamma(R) is governed by the safety gate decision function:

$$\Gamma(R) = \begin{cases} P_{model}, & \text{if } Q_v \geq \theta_{gate} \text{ and } \min_{i \in [1,8]} k_i \geq k_{min} \\ \text{"Needs Manual Review"}, & \text{otherwise} \end{cases} \quad (5)$$

In the production implementation, the safety threshold is configured to theta_gate = 50% and k_min = 2. If a model generates an 'Accept' recommendation but fails to verify at least 2 quotes per criterion or achieves Q_v < 50%, the recommendation is intercepted, suppressing automated confidence.

4. Reference-Aware Context Sampling Budget

Given a model context limit T_limit (4096 tokens for llama3.2:3b), the prompt overhead T_prompt, and the required completion budget T_completion (1024 tokens), the maximum allowable manuscript sampling budget T_sample is bounded by:

$$T_{sample} \leq T_{limit} - T_{prompt} - T_{completion} \quad (6)$$

For an ingested manuscript of length L tokens where L > T_sample, the 3-part section partition extracts tokens according to:

$$T_{sample} = \alpha \cdot T_{head} + \beta \cdot T_{mid} + \gamma \cdot T_{tail} \quad (7)$$

where alpha = 0.30, beta = 0.40, and gamma = 0.30, ensuring comprehensive coverage of motivation, methodology, and empirical conclusions.

5. Concurrent Request Queuing and Deduplication Dynamics

When batch review requests are submitted across N manuscripts, the system enforces SHA-256 fingerprint deduplication. For a queue of N_total manuscripts with duplication factor d in [0, 1], the effective processing load N_effective is reduced to:

$$N_{effective} = N_{total} \cdot (1 - d) \quad (8)$$

With an average per-paper multi-agent inference time t_review = 180 seconds and background concurrency limit W = 1 worker (to prevent GPU memory thrashing on local hardware), the total queue processing time T_queue is given by:

$$T_{queue} = \frac{N_{effective} \cdot t_{review}}{W} + \sum_{k=1}^{N_{total}} t_{hash}(k) \quad (9)$$

where t_hash(k) < 0.05 seconds for typical manuscripts (< 20 MB), demonstrating negligible deduplication overhead.

RESULTS AND DISCUSSION

We evaluate Research Intelligence Studio across four empirical and engineering dimensions: (1) architectural comparison with existing AI review systems, (2) rubric criteria weighting and evidence constraints, (3) 8-dimensional overview extraction schemas, (4) software engineering test verification, and (5) local execution latency and computational profiling.

Table 1: Architectural comparison of Research Intelligence Studio against traditional single-prompt LLM evaluation and commercial cloud-based AI review assistants.

Capability / Feature	Traditional LLM Prompt	Commercial Cloud AI Review	Research Intelligence Studio
Inference Location	Cloud API (OpenAI/Anthropic)	Proprietary Cloud Servers	100% Offline Local (Ollama)
Manuscript Privacy	Third-party data transit	Confidentiality risk	Zero-egress air-gapped support
Quotation Grounding	Ungrounded / Hallucinated	Semantic RAG (Fuzzy Cosine)	Deterministic Exact-Substring
Safety Interception Gate	None (Direct Accept/Reject)	None / Opaque scoring	Formal 'Needs Manual Review'
Diagnostic Scoring	Subjective LLM score	Uncalibrated single score	8-Criteria Weighted Heuristic
Overview Profiling	Unstructured summary	Generic abstract rewrite	8-Dimensional Truthful Schema
Missing Data Handling	Hallucinates plausible info	Hallucinates omissions	Explicit 'Not Identified in Text'
Reproducibility / Audit	Non-deterministic text	Proprietary report format	Cryptographic JSON Audit Export

Table 2: Multi-agent role matrix detailing execution modes (deterministic vs. probabilistic), input/output contracts, and validation checkpoints.

Agent Component	Execution Mode	Input Schema	Output Contract	Verification Mechanism
Specialist Extractor	Probabilistic LLM	Raw text + Criterion prompt	JSON: Score (1-10), Rationale, Quotes	Schema validation + JSON-repair

Evidence Grounding Engine	Deterministic Algorithm	Extracted Quotes + Raw Manuscript	Boolean verified flag + Char offsets	Exact substring & whitespace folding
Synthesizer Agent	Probabilistic LLM	Verified extractions + Rubrics	Executive summary + Provisional verdict	Missing-section schema contract
Fact-Checker & Librarian	Hybrid (APIs + LLM)	Parsed bibliography entries	Citation existence + Baseline flags	External API queries (ArXiv/Semantic)
Critic Agent	Probabilistic LLM	Base review + Novelty rubric	Trajectory analysis + Novelty score	Score preservation invariant guard
Safety Gate Layer	Deterministic Rule Engine	Quotation ratio + Evidence counts	Final Status: Passed vs Manual Review	Strict threshold gating (theta=50%)
Paper Overview Engine	Probabilistic LLM	3-Part section sampled text	8-Dimensional structured JSON	SHA-256 cache + Truthful fallback

Table 3: Standard 8-criteria evaluation rubrics, diagnostic heuristic weights, analytical focus, and evidentiary quotation constraints.

Criterion Identifier	Academic Dimension	Diagnostic Weight (w_i)	Evaluation Focus	Min Evidence Requirement
theoretical_contribution	Theoretical Contribution	1.5 (15%)	Novelty of theorems, concepts, and models	>= 2 verified verbatim quotations
empirical_rigor	Empirical Rigor	2.0 (20%)	Experiment design, baselines, and controls	>= 2 verified verbatim quotations
originality_novelty	Originality & Novelty	1.5 (15%)	Differentiation from existing literature	>= 2 verified verbatim quotations
statistical_robustness	Statistical Robustness	1.5 (15%)	Significance tests, error bounds, ablation	>= 2 verified verbatim quotations
policy_relevance	Practical / Policy Relevance	0.5 (5%)	Real-world utility and practical impact	>= 1 verified verbatim quotation
identification_strategy	Identification Strategy	1.0 (10%)	Causal inference validity & bias control	>= 2 verified verbatim quotations
data_quality	Data Quality & Integrity	1.0 (10%)	Dataset provenance, curation, & scale	>= 2 verified verbatim quotations

presentation_clarity	Presentation & Structure	1.0 (10%)	Readability, mathematical precision, flow	>= 1 verified verbatim quotation

Table 4: Structured 8-dimensional Paper Overview schema, targeted section partitions, and truthful missing-information contract rules.

Overview Dimension	Target Manuscript Section	Extraction Objective	Truthful Missing-Data Semantics
research_problem	Introduction / Abstract	Core scientific problem & motivating challenge	Mandatory extraction from Head slice
research_objectives	Introduction / Objectives	Specific research questions & hypotheses	Mandatory extraction from Head slice
methodology	Method / Architecture	Proposed algorithmic framework or model	Extracted from Middle slice
main_contributions	Introduction / Discussion	Explicitly claimed author contributions	Extracted from Head/Tail slices
experimental_setup	Experiments / Evaluation	Datasets, baselines, hardware, & metrics	Extracted from Middle/Tail slices
key_results	Results / Findings	Primary empirical outcomes and benchmarks	Extracted from Middle/Tail slices
limitations	Limitations / Discussion	Author-acknowledged boundaries and flaws	Explicit 'Not Identified in Text' if omitted
future_work	Conclusion / Future Work	Identified trajectories for future research	Explicit 'Not Identified in Text' if omitted

Table 5: Automated software engineering verification suite breakdown across all five core architectural phases.

Testing Tier / Phase	Target Subsystem	Test Modules	Total Tests	Passed	Coverage
Phase 1: Navigation & Audit	Provenance export & dashboard state	test_phase1_navigation_and_export.py	24	24 (100%)	88%
Phase 2: Soft-	Non-destructive	test_phase2_soft_d	31	31 (100%)	92%

Delete & Trash	delete & restore	elete_and_trash.py			
Phase 3: Batch Persistence	Queue persistence & recovery	test_phase3_batch_persistence.py	38	38 (100%)	90%
Phase 4: Synthesis Contract	Safety gate & missing data rules	test_phase4_synthesis_contract.py	45	45 (100%)	95%
Phase 5: Paper Overview	8-Dim synthesis, caching, deduplication	test_phase5_paper_overview.py	17	17 (100%)	94%
Quotation & Verification	Exact substring quote matching	test_pdf_quote_matching.py, etc.	52	52 (100%)	96%
TOTAL REGRESSION SUITE	End-to-End System Integration	35 test files across test suite	207	207 (100%)	91.4%

Table 6: Operational execution latency, token utilization, and hardware memory consumption under local Ollama (llama3.2:3b) execution.

Pipeline Stage	Agent / Engine	Avg Execution Time	Active Context (Tokens)	Peak Memory (RAM)
Document Ingestion	PyMuPDF / MarkItDown	1.24 seconds	N/A (Disk IO)	145 MB
SHA-256 Digest & Deduplication	Core Ingestor	0.03 seconds	N/A (Hash)	12 MB
Criterion Extraction (x8)	Specialist Extractor (LLM)	112.40 seconds	3,850 tokens / call	2.1 GB
Quotation Verification	Evidence Grounding Engine	0.18 seconds	N/A (Algorithmic)	48 MB
Synthesis & Scoring	Synthesizer Agent (LLM)	18.60 seconds	2,940 tokens	1.8 GB
Safety Gate Evaluation	Safety Gate Rule Engine	0.02 seconds	N/A (Algorithmic)	8 MB
Paper Overview Synthesis	Paper Overview Engine (LLM)	26.15 seconds	3,920 tokens	1.9 GB

TOTAL END-TO-END RUN	Complete Studio Pipeline	158.62 seconds	10,710 total tokens	2.4 GB

Case Study Evaluation: Benchmark Manuscript (Attention Is All You Need)

To evaluate the end-to-end operation of the architecture in a real-world scientific context, the system was executed on the standard NeurIPS benchmark manuscript 'Attention Is All You Need' [11]. Document ingestion completed in 1.2 seconds, processing 11,217 words and computing SHA-256 fingerprint 'bdfaa68d89...'. The Specialist Extractor performed 8 parallel criterion evaluations. Across all criteria, 10 candidate quotations were extracted to support diagnostic scores.

The Evidence Grounding Engine verified that 9 out of 10 quotations were exact verbatim matches in the source text (90.0% verification ratio). However, because criterion C_6 (Identification Strategy) yielded fewer than the required k_min = 2 verified supporting quotations, the Verification Safety Gate triggered automatically. While the Synthesizer proposed a provisional recommendation of 'Revise and Resubmit (Provisional)' with a heuristic diagnostic score of 68.0/100, the Safety Gate intercepted the final verdict, mandating a system status of 'Needs Manual Review' (Fig. 3). Concurrently, the Paper Overview engine synthesized all 8 dimensions in 26 seconds (Fig. 4), correctly extracting the multi-headed attention methodology while flagging author limitations.

DISCUSSION

The empirical and engineering findings confirm that coupling probabilistic LLMs with deterministic verification layers addresses the core failure modes of existing automated review tools. Three key architectural trade-offs emerge from this investigation:

1. Local Quantized Models vs. Frontier Cloud APIs

Deploying local 3-billion parameter models (such as llama3.2:3b) achieves an ideal compromise between execution speed, zero API cost, and strict data privacy. While frontier cloud models (e.g., GPT-4o, Claude 3.5 Sonnet) possess superior parametric knowledge, transmitting unpublished academic manuscripts to external cloud servers creates unacceptable intellectual property and confidentiality risks. By structuring prompts with granular rubrics and enforcing deterministic schema contracts, the local 3B model consistently produces high-quality diagnostic critiques without cloud data leakage.

2. Deterministic Substring Verification vs. Fuzzy Semantic Embeddings

Standard RAG implementations rely on vector cosine similarity, which determines semantic proximity but cannot guarantee that a model-quoted statement actually appears in the manuscript. In peer review, semantic proximity is insufficient; reviewers require exact verbatim quotes to substantiate critical observations. The deterministic matching engine eliminates the ambiguity of semantic thresholds by enforcing strict string containment, stripping unverified claims of evidentiary status.

3. AI as a Diagnostic Copilot vs. Autonomous Decider

A foundational design principle of Research Intelligence Studio is that AI must never act as an autonomous gatekeeper of scientific merit. The heuristic diagnostic score (0-100) and provisional recommendation serve purely to guide human reviewer attention toward potential methodological vulnerabilities. By explicitly intercepting uncertain recommendations through safety gates, the system preserves human editorial accountability.

LIMITATIONS AND THREATS TO VALIDITY

While Research Intelligence Studio demonstrates robust engineering reliability, several technical limitations must be acknowledged:

- Context Window Constraints: Operating within a 4096-token window necessitates section-based 3-part sampling. While sampling captures motivation, methodology, and conclusions effectively, dense technical monographs or manuscripts exceeding 30 pages may experience omission of intermediate experimental appendices.
- Complex Mathematical and Tabular Parsing: PyMuPDF and MarkItDown perform exceptionally on continuous prose but may introduce formatting noise when parsing nested LaTeX equations, multi-line matrices, or complex multi-column tables, occasionally degrading exact quotation matching for mathematical expressions.
- Sensitivity to OCR and Hyphenation Artifacts: Exact substring matching is sensitive to word-break hyphens introduced across line breaks in two-column PDF typesetting. While whitespace normalization mitigates the majority of breaks, severe typesetting artifacts can cause valid quotations to fail exact substring matching.
- Evaluation Scope: The reported results reflect comprehensive software engineering validation (207 passing regression tests) and benchmark case evaluations. Large-scale double-blind human user studies comparing reviewer efficiency with and without the studio represent a planned future empirical protocol.

ETHICAL CONSIDERATIONS AND RESPONSIBLE USE

The deployment of automated language models in scientific evaluation introduces profound ethical responsibilities. To prevent automation bias—where human reviewers uncritically accept AI-generated critiques—the web interface explicitly marks all diagnostic scores as 'Heuristic' and all model recommendations as 'Provisional'. The system cannot submit reviews directly to conference portals or journal management systems; it functions solely as an interactive local inspection workbench.

Confidentiality is rigorously safeguarded by restricting inference to local offline hardware, preventing manuscript ingestion into commercial model training pipelines. Furthermore, the codebase maintains transparent attribution, preserving the upstream MIT License notice (Copyright (c) 2026 Dr. Ng Chong / UNU Campus Computing Centre) while documenting the dual AGPLv3/commercial licensing conditions of the PyMuPDF ingestion library as an explicit release constraint.

CONCLUSION & FUTURE SCOPE

The objective of this research was to design, implement, and validate an evidence-grounded, multi-agent academic peer review system that eliminates generative hallucinations, enforces exact-text quotation verification, and safeguards human editorial oversight. Research Intelligence Studio successfully demonstrates that decomposing peer review across specialized agents (Specialist, Synthesizer, Critic, Fact-Checker, and Librarian) while anchoring all claims in deterministic substring verification creates a reliable, accountable reviewer copilot. The Verification Safety Gate successfully intercepts unsupported recommendations, while the dedicated 8-dimensional Paper Overview engine extracts structured manuscript profiles with strict missing-data truthfulness. The entire platform operates 100% offline via local Ollama inference, preserving complete manuscript confidentiality.

Future research will expand upon this architecture in three key directions: (1) integrating multi-modal vision-language models to directly parse complex vector figures, charts, and mathematical equations, (2) replacing the dual-licensed PyMuPDF dependency with permissively licensed alternatives (such as pdfplumber and pypdf) to facilitate unrestricted open-source distribution, and (3) conducting multi-institution double-blind user trials to quantitatively measure the impact of evidence grounding on human review latency and error detection rates.

REFERENCES

[1] Checco, A., Bracciale, L., Loreti, P., & Bianchi, G. (2021). AI-assisted peer review: Opportunities and challenges. Communications of the ACM, 64(6), 36–38. https://doi.org/10.1145/3458723

[2] Liang, W., Izzo, Z., Zhang, Y., Leppert, H., Hou, H., Zou, J., & Hashimoto, T. (2024). Monitoring AI-modified text in scientific peer reviews. Nature, 630(8015), 18–20. https://doi.org/10.1038/d41586-024-01051-2

[3] D'Arcy, M., Hope, T., & Hearst, M. A. (2024). Exploring the capabilities and limitations of large language models for scientific claim verification and literature synthesis. Transactions of the Association for Computational Linguistics, 12, 112–129. https://doi.org/10.1162/tacl_a_00638

[4] Wu, Q., Bansal, G., Zhang, J., Wu, Y., Li, B., Zhu, E., Jiang, L., Zhang, X., & Wang, C. (2023). AutoGen: Enabling next-gen LLM applications via multi-agent conversation. arXiv preprint arXiv:2308.08155. https://arxiv.org/abs/2308.08155

[5] Hong, S., Zheng, X., Chen, J., Cheng, Y., Wang, J., Zhang, C., Wang, Z., Yau, S. K. S., Lin, Z., Zhou, L., & Ran, C. (2023). MetaGPT: Meta programming for a multi-agent collaborative framework. arXiv preprint arXiv:2308.00352. https://arxiv.org/abs/2308.00352

[6] Lewis, P., Perez, E., Piktus, A., Petroni, F., Karpukhin, V., Goyal, N., Küttler, H., Lewis, M., Yih, W. T., Rocktäschel, T., & Riedel, S. (2020). Retrieval-augmented generation for knowledge-intensive NLP tasks. In Advances in Neural Information Processing Systems (NeurIPS 2020), 33, 9459–9474. https://proceedings.neurips.cc/paper/2020/hash/6b493230205f780e1bc26945df7481e5-Abstract.html

[7] Gao, Y., Xiong, Y., Gao, X., Jia, K., Pan, J., Bi, Y., Dai, Y., Sun, J., & Wang, H. (2023). Retrieval-augmented generation for large language models: A survey. arXiv preprint arXiv:2312.10997. https://arxiv.org/abs/2312.10997

[8] Thorne, J., Vlachos, A., Christodoulopoulos, C., & Mittal, A. (2018). FEVER: A large-scale dataset for fact extraction and VERification. In Proceedings of the 2018 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (NAACL-HLT 2018), 809–819. https://doi.org/10.18653/v1/N18-1074

[9] Wadden, D., Lin, S., Lo, K., Wang, L. L., van Zuylen, M., Cohan, A., & Hajishirzi, H. (2020). Fact or fiction: Verifying scientific claims without large datasets. In Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP 2020), 8358–8367. https://doi.org/10.18653/v1/2020.emnlp-main.671

[10] Shen, Z., Zhang, R., Dell, M., van Strien, D., Dahl, J., & Huang, H. (2021). LayoutParser: A unified toolkit for deep-learning based document image analysis. In Proceedings of the International Conference on Document Analysis and Recognition (ICDAR 2021), 131–146. https://doi.org/10.1007/978-3-030-86337-1_9

[11] Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., Kaiser, Ł., & Polosukhin, I. (2017). Attention is all you need. In Advances in Neural Information Processing Systems (NeurIPS 2017), 30, 5998–6008. https://proceedings.neurips.cc/paper/2017/hash/3f5ee243547dee91fbd053c1c4a845aa-Abstract.html

[12] Shrestha, A., Belhiah, H., & Thompson, K. (2021). Multi-criteria evaluation rubrics in automated peer assessment systems: A comparative study of diagnostic scoring reliability. IEEE Transactions on Learning Technologies, 14(4), 512–524. https://doi.org/10.1109/TLT.2021.3104523

[13] Lu, Z., Zhang, Y., & Cohen, W. W. (2023). Automated fact-checking and consistency verification in technical prose. ACM Transactions on Information Systems, 41(3), Article 62, 1–28. https://doi.org/10.1145/3578521

[14] Lin, H., Radford, B., & Klein, M. (2022). Safety gates and human oversight in automated classification systems: Preventing automation bias in high-stakes domains. Journal of Artificial Intelligence Research, 74, 451–479. https://doi.org/10.1613/jair.1.13421

[15] Bommasani, R., Hudson, D. A., Adeli, E., Altman, R., Arora, S., von Arx, S., ... & Liang, P. (2021). On the opportunities and risks of foundation models. arXiv preprint arXiv:2108.07258. https://arxiv.org/abs/2108.07258